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Agronomic Validation Dossier

Validation of the JSON Structure Representing a Full Crop Cycle

Supported by Innosuisse

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Background IP : Jan Affolter ITS (CHE-104.801.264), owner of the CompaniX project.

Foreword

Modern agriculture requires tools capable of describing agronomic reality with precision, without unnecessary complexity and without interpretation. In a context where farms seek to understand performance gaps between parcels that appear similar, the ability to document **what the crop actually experienced** becomes essential.

The CIF – Cycle Invariant File meets this requirement. It is a standardized format designed to capture the agronomic invariants of a crop cycle, organized by physiological stages. The CIF does not model, predict, or simulate: it **observes**, **structures**, and **documents** the cycle as it unfolded.

Version 2.0 of the CIF introduces a more detailed description of management interventions, a clear separation between cycle invariants and fine soil analyses, and a strengthened structuring of environmental and biotic factors. This evolution aims to ensure rigorous comparability between parcels, seasons, and environments, while remaining simple to fill in and robust to analyze.

It is important to note that **fine soil analyses** (NPK, organic matter, CEC, pH, granulometry, trace elements, etc.) **are not included in the CIF**. They are performed by specialized laboratories, stored in a separate system, and used by CompaniX analysis modules for multi-parcel comparison. The CIF focuses exclusively on **cycle invariants** in order to remain lightweight, readable, universal, and directly comparable across seasons.

This document presents the complete structure of CIF 2.0 and seeks formal agronomic validation. It is intended for cooperatives, technical institutes, agronomists, and advisors, and aims to establish a common methodological foundation for the objective analysis of crop performance.

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1 Purpose of the Document

This dossier seeks **formal agronomic validation** of the **CIF – Cycle Invariant File JSON structure**. The objective is to confirm that this format is **scientifically grounded**, **physiologically coherent**, and **sufficient to explain yield differences** between parcels that share:

- **the same soil type**,
- **the same climate**,
- **the same cultivated variety**.

In such a homogeneous context, the CIF must make it possible to identify the **factors truly responsible for performance gaps**: **management**, **stress**, **variability**, **water dynamics**, and **biotic pressure**.

The CIF summarizes the crop cycle through **agronomic invariants** that reflect the **conditions actually experienced by the crop**.

This document is intended for **cooperatives**, **agronomists**, **applied research institutes**, and **field advisors**.

2 Role and Scope of the CIF

The CIF (Crop Invariants File) is the **documentary standard** used by CompaniX to describe, in a structured and comparable way, the **agronomic conditions actually experienced by a crop** during a production cycle. It is a **factual, neutral, and operational file**, designed to enable **comparative performance analysis** between parcels, seasons, and environments.

The CIF does **not** model the crop: it documents the **invariants that directly influence the observed yield**.

2.1 What the CIF captures

The CIF captures **only agronomic invariants**, meaning factors that are:

- **measurable**,
- **comparable**,
- **reproducible**,
- **and directly linked to crop performance**.

These elements are organized by **physiological stages**, enabling a coherent reading of the cycle regardless of calendar or geographic location.

The CIF includes:

a. Agro-environmental conditions experienced by the crop

- soil moisture and temperature,
- intra-parcel variability,
- canopy microclimate,
- thermal stresses (frost, heat),
- disease and weed pressure.

b. Management interventions

Each intervention is described individually:

- product applied (entered by the farmer),
- date and dose,
- active substance, mode of action, formulation, and concentration (**automatically derived by country**),
- purpose of the treatment,
- timing and intensity indices (**automatically calculated**).

c. Major constraints

- frost at establishment,

- water stress at flowering,
- excess water,
- heat shock.

d. Observed yield

The CIF records the **actual yield measured at harvest**.

2.2 What the CIF does not capture

The CIF does **not** contain **fine soil analyses** (NPK, organic matter, CEC, pH, granulometry, trace elements, etc.). These data:

- are produced by specialized laboratories,
- are stored in a separate system,
- and are integrated into comparative analyses through dedicated CompaniX modules.

This separation ensures:

- **a lightweight and readable CIF**,
- **consistent comparisons between cycles**,
- **the integrity of laboratory-derived data**.

The CIF also does **not** contain raw time-series data (weather, sensors), but only the **agronomically relevant invariants derived from them**.

2.3 Role of the CIF within the CompaniX system

The CIF is the **backbone of CompaniX agronomic analysis**. It enables:

- **rigorous comparison between cycles**,
- **identification of the factors explaining yield differences**,
- **linking interventions, experienced conditions, and final outcomes**,
- **construction of a coherent multi-year history**.

The CIF is an **open standard**, designed to be **simple to fill in**, **robust to analyze**, and **universally applicable**.

3 General Structure of the JSON

The JSON describes a crop cycle through:

- **5 physiological stages** (establishment → vegetative → flowering → grain filling → maturation),
- **invariants** for soil, climate, variability, biotic stress, and management,
- **normalized indices** (0–1), min/max values, amplitudes, CV,
- a **compact size of 7–8 kB**.

A structure that is intentionally **simple**, **physiological**, and **comparable** across parcels and seasons.

4 Physiological Stages

The stages used are **standard and agronomically relevant**:

- **Establishment** — development of the root system
- **Vegetative** — biomass accumulation
- **Flowering** — reproductive initiation
- **Grain Filling** — deposition of reserves
- **Maturation** — senescence

→ **Fully aligned with crop physiology and suitable for aggregating agronomic signals.**

5 Soil Invariants

Includes: **minimum soil moisture / amplitude**, **minimum soil temperature / amplitude**, **EC variability**, **compaction**, **micro-topography**. Determinants of: **root development**, **water availability**, **nutrition**, **resilience**, **heterogeneity**.

→ A **complete set** for explaining **pedological effects on yield**.

6 Climatic Invariants

Includes: **minimum/maximum temperatures**, **humidity**, **VPD**, **heat-day count**, **frost-day count**. Impacts: **phenology**, **stress**, **diseases**, **biomass**, **grain filling**.

→ **Essential climatic variables, without redundancy.**

7 Variability Invariants

Includes: **CV of moisture**, **CV of temperature**, **CV of EC**, **uniformity indices**. Influences: **emergence**, **tillering**, **disease propagation**, **uniformity of maturity**, **harvestability**.

→ A **correct representation of spatial variability**.

8 Biotic Stress Invariants

Includes: **disease pressure**, **disease damage**, **lodging**, **weed pressure**.

→ The **main biotic factors** responsible for **yield reduction**.

9 Management Invariants

Management is represented through **a list of individual interventions**, each describing:

- **product**,
- **date**,
- **dose**,
- **active substance**,
- **mode of action**,
- **formulation**,
- **concentration**,
- **timing index**,
- **intensity index**,
- **purpose of the treatment**.

→ This structure is **precise**, **explanatory**, and **faithful to agronomic reality**.

10 Completeness

The five families of yield determinants are covered:

- **Soil**
- **Climate**
- **Variability**
- **Biotic stress**
- **Management** (multi-intervention format)

→ **No critical determinant is missing.**
→ **A structure suited to full-cycle analysis.**

11 Previous Crop and Irrigation

Previous Crop

Represented through the **CIF from the previous season (CIF N-1)**, which captures: water dynamics, compaction, residual fertility, diseases, weeds, variability, stage-specific stresses, and establishment quality.

→ **More precise than a simple "previous_crop" field.**

Irrigation

No "irrigated: yes/no" field. Actual water conditions are measured directly: **minimum moisture / amplitude**, **VPD**, **soil temperature**, **wetness**, **stage-specific water stress**.

→ **The type of infrastructure is irrelevant; only the physiological response matters.**

12 Why the CIF Represents a Major Advancement

The CIF – Cycle Invariant File constitutes a **structural breakthrough** in the way crop cycles are documented, analyzed, and leveraged. It transforms agricultural traceability — **entered once by the farmer** — into a set of **enriched, scientifically usable, and institutionally robust data**.

The CIF converts:

- **mandatory traceability**, → into **enriched data** (active substances, modes of action, physiological indices), → into **structured agronomic analysis**, organized by stages, → into **rigorous, reproducible multi-parcel comparison**, → into **official reporting**, compliant with regulatory requirements, → into **strategic decision-making**, improving performance and resilience.

Beyond this value chain, the CIF also enables validation of the **appropriateness of the management schedule**, cycle after cycle.
By integrating **timing and intensity indices** and structuring interventions by **physiological stage**, it becomes possible to evaluate the **synchronization between management decisions and the actual climatic conditions experienced**.
This approach does not rely on the notion of error, but on **physiological coherence**, **responsiveness to variability**, and **adaptation of practices** from one season to the next.

Thanks to this capacity for continuous analysis, the CIF becomes an essential tool to:

- understand the **real impact** of practices,
- strengthen the **robustness** of management strategies,
- document the **quality of interventions**,
- support **sustainability**, **auditing**, and **compliance** processes.

The CIF thus establishes itself as a **modern operational standard**, both **lightweight for the farmer** and **powerful for institutions**, capable of supporting agronomic transition and improving decision quality at every scale.

The CIF can become the **single point of data entry** for the farmer.
CompaniX then automatically generates all **formalized reports** required by agricultural authorities.

13 Value of the CIF for Farmers

The CIF (Cycle Invariant File) is designed to **simplify the farmer's work**. It adds **no administrative burden**: it uses only the information the farmer already records in the traceability register (products, dates, doses, observations, yield).

13.1 Why the CIF is useful for farmers

13.1.1 A single data-entry point

The CIF can become the **only place where the farmer enters operational traceability**. A single entry, in one place, with the same information as usual. CompaniX then automatically generates all official reports.

From the CIF, CompaniX automatically produces:

- **regulatory reports**,
- **documents for audits**,
- **summaries requested by cooperatives**,
- **formalized exports for agricultural authorities**.

→ **The farmer enters data once.**
→ **CompaniX handles everything else.**

13.1.2 A clear understanding of what influenced yield

The CIF highlights:

- the **alignment between the management schedule and climatic conditions**,
- **timing variations** between parcels or seasons,
- the **actual crop response** to soil and climate conditions.

When needed, the CIF helps adjust the management schedule based on the **real conditions experienced** in the field.

13.1.3 Objective comparison between parcels

Same soil, same variety, same weather… yet different yields. The CIF makes it possible to understand **why**.

13.1.4 Traceability that creates value

The CIF can be used for:

- **quality programs**,
- **certifications**,

- **audits**,
- **administrative applications**.

13.1.5 No additional work

The system automatically enriches the data (active substance, mode of action, indices, etc.). The farmer does **nothing more** than what they already do.

14 Value of the CIF for Agricultural Authorities

The CIF is a **standardized, neutral, and verifiable format** that documents a crop cycle in a coherent, comparable, and scientifically grounded manner. It relies exclusively on the data that farmers are already required to record in their traceability register, **without creating any additional administrative burden**.

14.1 Why the CIF is relevant for agricultural authorities

14.1.1 A single entry point for farm traceability

The CIF can serve as a **unified data-entry support** for farmers. All mandatory information is centralized in a **structured and harmonized format**.

From the CIF, CompaniX automatically generates:

- **formalized reports for authorities**,
- **audit documents**,
- **standardized exports**,
- **summaries of crop management practices**.

This ensures **methodological consistency** between traceability, agronomic analysis, and reporting.

14.1.2 A harmonized and comparable format

The CIF facilitates:

- **regional or national consolidation**,
- **analysis of farming practices**,
- **monitoring of inputs**,
- **sustainability programs**,
- **agroecological transition initiatives**.

14.1.3 Enriched and verifiable traceability

Interventions are automatically completed with:

- **active substance**,
- **mode of action**,
- **formulation**,
- **concentration**,
- **timing and intensity indices**.

This strengthens the **reliability** and **transparency** of the data.

14.1.4 A physiological view of the crop cycle

The CIF structures information by **physiological stages**, enabling a more agronomically relevant interpretation than simple calendar dates.

14.1.5 Compatibility with public policies

The CIF supports:

- **input-reduction strategies**,
- **climate-resilience programs**,
- **certification schemes**,
- **practice-monitoring systems**.

15 List of Acronyms Used

Acronym	Meaning	Description
CIF	**Cycle Invariant File**	CompaniX JSON file representing a crop cycle through agronomic invariants.
CV	**Coefficient of Variation**	Measure of spatial variability (relative differences in moisture, temperature, EC, etc.).
EC	**Electrical Conductivity**	Indicator of soil salinity, structure, texture, and spatial variability.
VPD	**Vapor Pressure Deficit**	Key indicator of water stress and transpiration.
N-1	**Previous Crop Cycle**	Represented by the CIF from the previous season, capturing agronomic legacy.

16 JSON CIF File – Version for production

The **actual JSON file** that conforms to the CIP specification.
A CIF contains:

- metadata (crop, variety, year, plot)
- five physiological stages
- soil, climate, variability, pathology, and management invariants
- variety traits
- yield and constraints

A CIF is typically ~7-8 kB, making it sovereign, frugal, and portable.


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{
  "id_cycle": "P-12-wheat-2025",
  "plot_id": "P-12",
  "crop": "wheat",
  "variety": "Apache",
  "year": 2025,

  "variety_traits": {
    "earliness_index": 0.6,
    "drought_tolerance_index": 0.7,
    "disease_resistance_index": 0.8,
    "lodging_resistance_index": 0.9
  },

  "sowing_date": "2025-03-18",
  "harvest_date": "2025-08-12",

  "stages": [
    {
      "name": "establishment",
      "start_date": "2025-03-18",
      "end_date": "2025-04-02",

      "functional_soil": {
        "moisture_min": 18.4,
        "moisture_amplitude": 11.2,
        "soil_temperature_min": 8.1,
        "soil_temperature_amplitude": 4.0
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      "local_climate": {
        "air_temperature_min": 3.2,
        "air_temperature_amplitude": 12.5,
```

```
        "air_humidity_min": 55.0,
        "air_humidity_amplitude": 20.0,
        "heat_days": 0,
        "frost_days": 2
      },

      "soil_variation": {
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        "temperature_cv": 0.12,
        "ec_cv": 0.17,
        "compaction_index": 0.22,
        "texture_variability_index": 0.15,
        "microtopography_index": 0.11
      },

      "canopy_microclimate": {
        "vpd_min": 0.4,
        "vpd_amplitude": 1.2,
        "canopy_wetness_events": 4
      },

      "management": [
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          "product_name": "Roundup Max",
          "application_date": "2025-03-15T10:00",
          "active_substance": "glyphosate",
          "mode_of_action": "EPSPS_inhibitor",
          "concentration_g_l": 450,
          "formulation": "SL",
          "dose_applied": 2.0,
          "timing_index": 0.95,
          "intensity_index": 0.3,
          "treatment_goal": "pre_sowing_burndown"
        }
      ],

      "pathology": {
        "disease_pressure_index": 0.1,
        "disease_damage_index": 0.0,
        "lodging_index": 0.0,
        "weed_pressure_index": 0.3
      }
    },

    {
      "name": "vegetative",
      "start_date": "2025-04-03",
      "end_date": "2025-05-14",

      "functional_soil": {
```

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        "moisture_min": 16.1,
        "moisture_amplitude": 13.4,
        "soil_temperature_min": 10.2,
        "soil_temperature_amplitude": 5.1
      },

      "local_climate": {
        "air_temperature_min": 6.4,
        "air_temperature_amplitude": 18.2,
        "air_humidity_min": 48.0,
        "air_humidity_amplitude": 22.0,
        "heat_days": 1,
        "frost_days": 0
      },

      "soil_variation": {
        "moisture_cv": 0.20,
        "temperature_cv": 0.14,
        "ec_cv": 0.18,
        "compaction_index": 0.24,
        "texture_variability_index": 0.16,
        "microtopography_index": 0.12
      },

      "canopy_microclimate": {
        "vpd_min": 0.5,
        "vpd_amplitude": 1.4,
        "canopy_wetness_events": 6
      },

      "management": [
        {
          "product_name": "Moddus",
          "application_date": "2025-04-20T09:30",
          "active_substance": "trinexapac-ethyl",
          "mode_of_action": "growth_regulator",
          "concentration_g_l": 250,
          "formulation": "ME",
          "dose_applied": 0.3,
          "timing_index": 0.78,
          "intensity_index": 0.4,
          "treatment_goal": "lodging_prevention"
        },
        {
          "product_name": "Axial",
          "application_date": "2025-04-25T14:10",
          "active_substance": "pinoxaden",
          "mode_of_action": "ACCase_inhibitor",
          "concentration_g_l": 45,
          "formulation": "EC",
```

```
        "dose_applied": 0.9,
        "timing_index": 0.81,
        "intensity_index": 0.2,
        "treatment_goal": "grass_weed_control"
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    ],

    "pathology": {
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      "disease_damage_index": 0.05,
      "lodging_index": 0.0,
      "weed_pressure_index": 0.2
    }
  },

  {
    "name": "flowering",
    "start_date": "2025-05-15",
    "end_date": "2025-06-02",

    "functional_soil": {
      "moisture_min": 14.2,
      "moisture_amplitude": 14.3,
      "soil_temperature_min": 12.1,
      "soil_temperature_amplitude": 6.1
    },

    "local_climate": {
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      "air_temperature_amplitude": 21.2,
      "air_humidity_min": 42.0,
      "air_humidity_amplitude": 18.0,
      "heat_days": 4,
      "frost_days": 0
    },

    "soil_variation": {
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      "temperature_cv": 0.15,
      "ec_cv": 0.19,
      "compaction_index": 0.25,
      "texture_variability_index": 0.17,
      "microtopography_index": 0.13
    },

    "canopy_microclimate": {
      "vpd_min": 0.6,
      "vpd_amplitude": 1.6,
      "canopy_wetness_events": 3
    },
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    "management": [
      {
        "product_name": "Amistar",
        "application_date": "2025-05-18T08:30",
        "active_substance": "azoxystrobin",
        "mode_of_action": "QoI",
        "concentration_g_l": 250,
        "formulation": "SC",
        "dose_applied": 0.8,
        "timing_index": 0.82,
        "intensity_index": 0.6,
        "treatment_goal": "disease_control"
      },
      {
        "product_name": "Tebuconazole 250",
        "application_date": "2025-05-22T07:45",
        "active_substance": "tebuconazole",
        "mode_of_action": "DMI",
        "concentration_g_l": 250,
        "formulation": "EC",
        "dose_applied": 0.5,
        "timing_index": 0.88,
        "intensity_index": 0.5,
        "treatment_goal": "fusarium_prevention"
      }
    ],

    "pathology": {
      "disease_pressure_index": 0.4,
      "disease_damage_index": 0.1,
      "lodging_index": 0.0,
      "weed_pressure_index": 0.1
    }
  },

  {
    "name": "grain_filling",
    "start_date": "2025-06-03",
    "end_date": "2025-07-10",

    "functional_soil": {
      "moisture_min": 12.5,
      "moisture_amplitude": 10.8,
      "soil_temperature_min": 14.0,
      "soil_temperature_amplitude": 7.2
    },

    "local_climate": {
      "air_temperature_min": 10.1,
```

```
      "air_temperature_amplitude": 24.3,
      "air_humidity_min": 38.0,
      "air_humidity_amplitude": 20.0,
      "heat_days": 6,
      "frost_days": 0
    },

    "soil_variation": {
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      "temperature_cv": 0.13,
      "ec_cv": 0.16,
      "compaction_index": 0.21,
      "texture_variability_index": 0.14,
      "microtopography_index": 0.10
    },

    "canopy_microclimate": {
      "vpd_min": 0.7,
      "vpd_amplitude": 1.8,
      "canopy_wetness_events": 2
    },

    "management": [
      {
        "product_name": "Folpet 500",
        "application_date": "2025-06-05T09:00",
        "active_substance": "folpet",
        "mode_of_action": "multi_site",
        "concentration_g_l": 500,
        "formulation": "SC",
        "dose_applied": 1.5,
        "timing_index": 0.75,
        "intensity_index": 0.3,
        "treatment_goal": "late_disease_protection"
      }
    ],

    "pathology": {
      "disease_pressure_index": 0.3,
      "disease_damage_index": 0.15,
      "lodging_index": 0.1,
      "weed_pressure_index": 0.1
    }
  },

  {
    "name": "maturation",
    "start_date": "2025-07-11",
    "end_date": "2025-08-12",
```

```
    "functional_soil": {
      "moisture_min": 10.4,
      "moisture_amplitude": 9.1,
      "soil_temperature_min": 16.2,
      "soil_temperature_amplitude": 8.0
    },

    "local_climate": {
      "air_temperature_min": 12.4,
      "air_temperature_amplitude": 26.1,
      "air_humidity_min": 35.0,
      "air_humidity_amplitude": 17.0,
      "heat_days": 8,
      "frost_days": 0
    },

    "soil_variation": {
      "moisture_cv": 0.17,
      "temperature_cv": 0.11,
      "ec_cv": 0.15,
      "compaction_index": 0.20,
      "texture_variability_index": 0.13,
      "microtopography_index": 0.09
    },

    "canopy_microclimate": {
      "vpd_min": 0.8,
      "vpd_amplitude": 2.0,
      "canopy_wetness_events": 1
    },

    "management": [
      {
        "product_name": "Magnesium Nitrate",
        "application_date": "2025-07-15T06:50",
        "active_substance": "magnesium_nitrate",
        "mode_of_action": "nutrient",
        "concentration_g_l": 400,
        "formulation": "SL",
        "dose_applied": 3.0,
        "timing_index": 0.65,
        "intensity_index": 0.2,
        "treatment_goal": "grain_quality_boost"
      }
    ],

    "pathology": {
      "disease_pressure_index": 0.2,
      "disease_damage_index": 0.1,
      "lodging_index": 0.2,
```

```
        "weed_pressure_index": 0.1
      }
    }
  ],

  "yield": 6.8,
  "constraints": ["frost_at_establishment", "water_stress_flowering"],
  "soil_fao": "CMx"
}
```

Project Ownership

Supported by: Innosuisse

Owner & Founder: Jan Affolter — jan.affolter@gmail.com

Intellectual Property Holder: Jan Affolter ITS (CHE-104.801.264)

Location: Lausanne, Switzerland